

Cost of Living Report

UN - Department of Economic and Social Affairs

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Dataset

Cost of living is the amount of money required to maintain a certain standard of living, encompassing:

1

Restaurant

Meals, McDonald's, Drinks

2

Market

Diaries, Fruits, Meats, Vegetables, Drinks, Wines

3

Transportation

Local transport tickets, Taxi, Car

4

Utilities

Electrics, Waters, Internets

5

Sport & Leisure

Gym, Tennis, Cinemas

6

Cloth

Jeans, Dresses, Shoes

7

Accommodation

Renting apartments, Buying houses

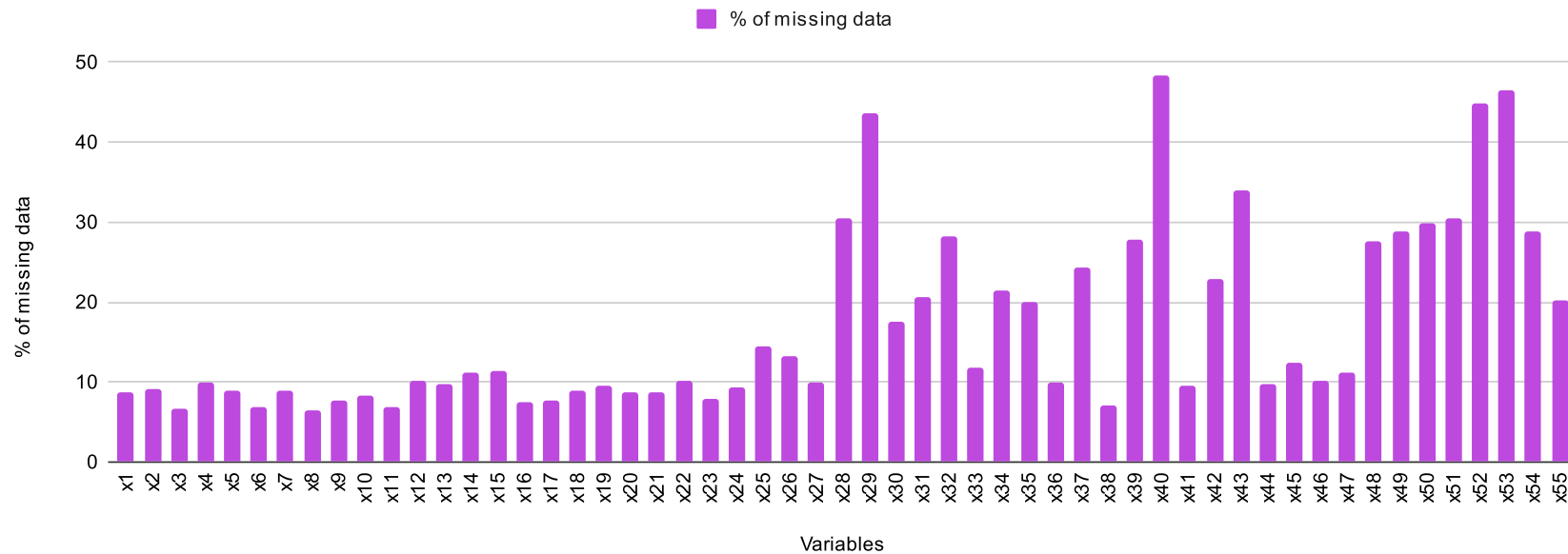
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Financial

Salaries, Mortgage rate

Data Cleaning

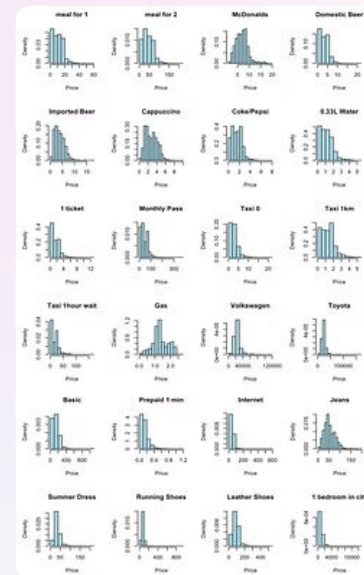
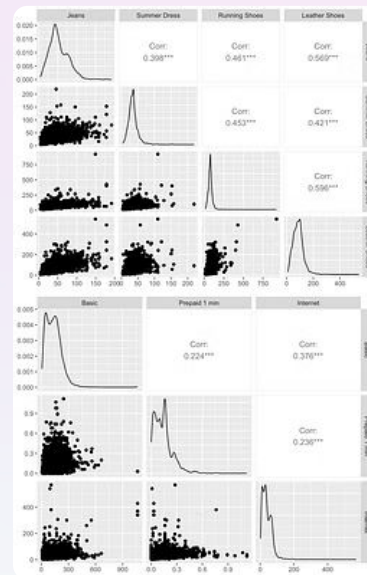
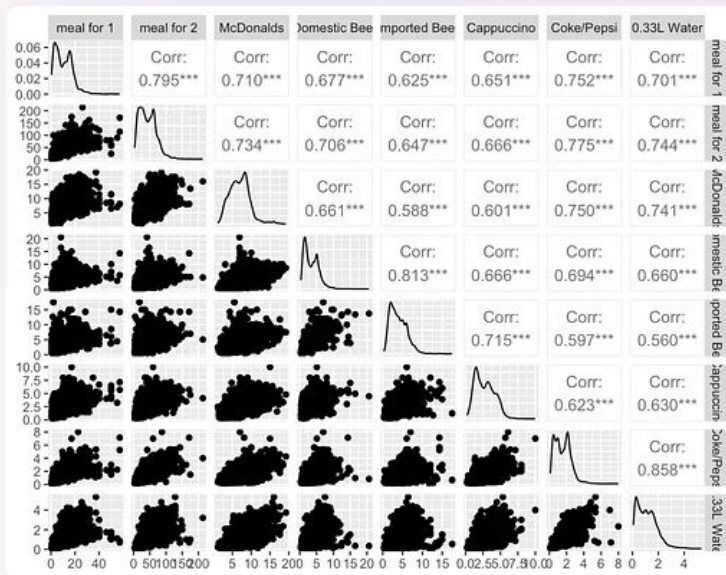
Percentage of Missing Data in Each Variable



- Low-populated countries and city-states, such as **Greenland** or **San Marino**, have limited data availability.
- Missing data was imputed at the lowest granularity (city) with the relative country mean.

How do living costs compare globally?

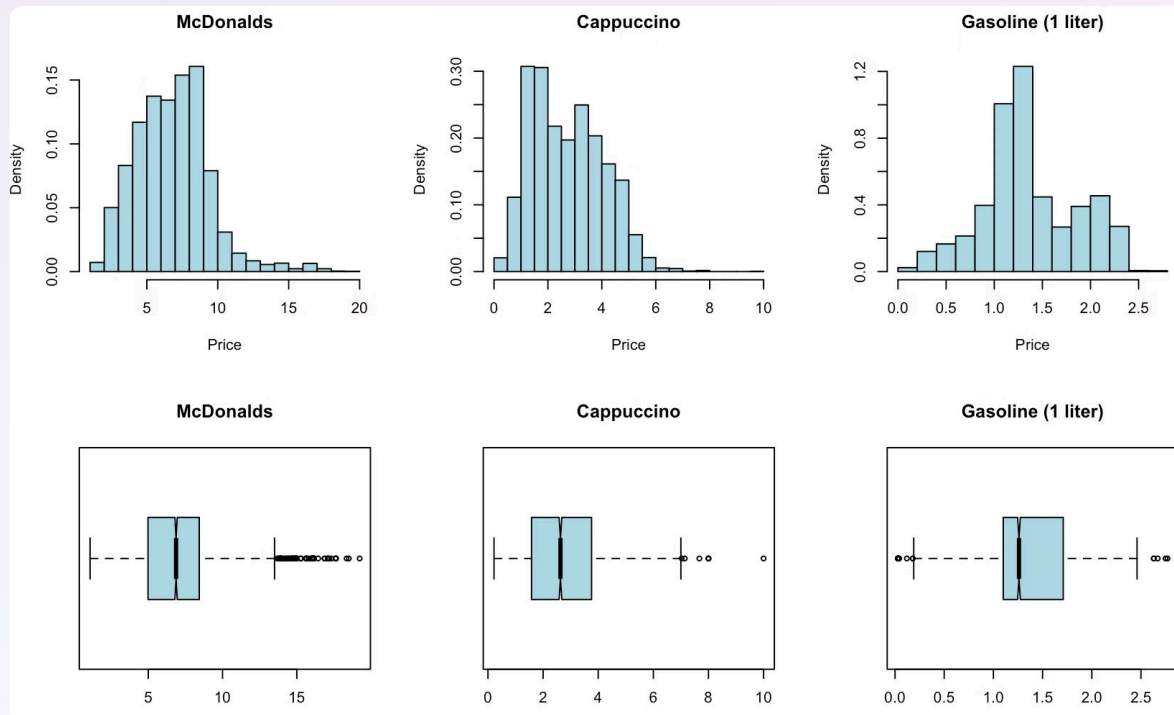
With ggpairs and histograms, most of the factors are extremely right-skewed distribution.



How do living costs compare globally?

Close to normal distribution :

- **McDonalds** : Global brand with standardized products and pricing strategy.
- **Cappuccino & Gasoline** :
 - Globally traded commodity produced and consumed in numerous countries worldwide.
 - Relatively balanced supply-demand relationship.



Cost of Living Index

What factors contribute to variations in the cost-of-living index (COLI) across cities?

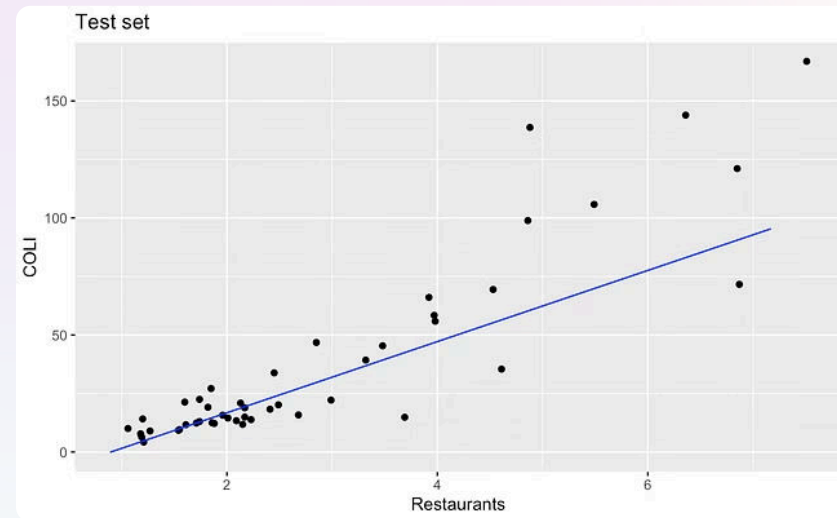
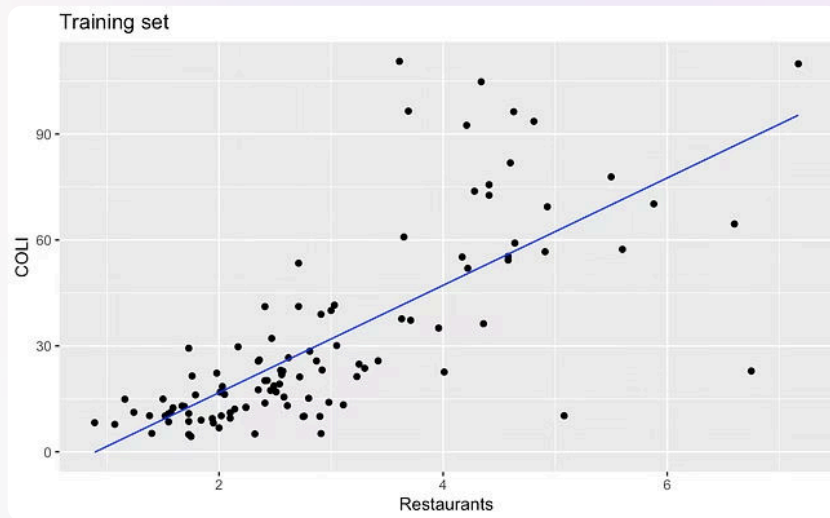
The Cost of Living Index (COLI) is a measure designed to quantify the relative cost of living globally.

Most Important Categories:

- Meal (20%)
- Market goods (20%)
- Monthly Rent (20%)

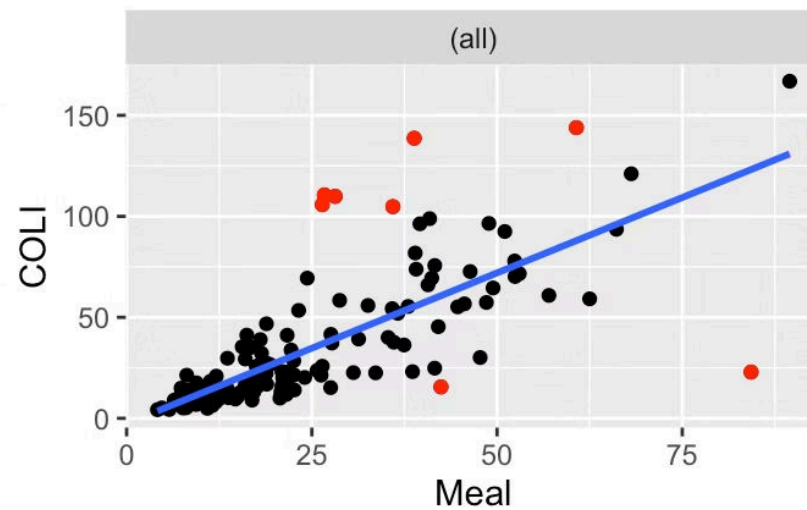
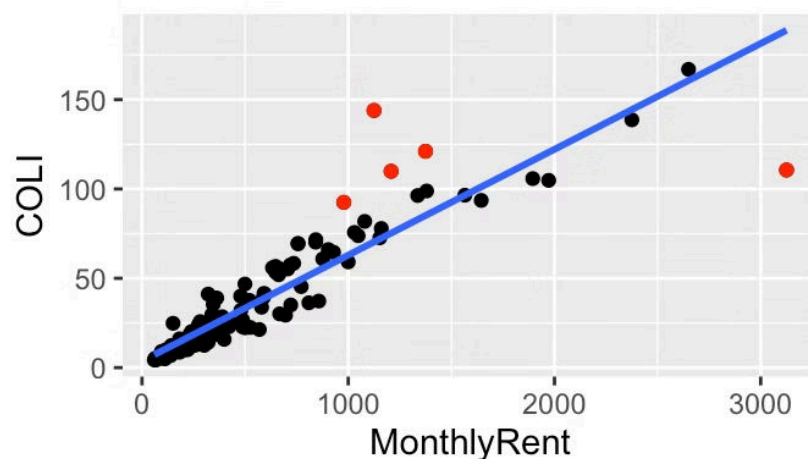
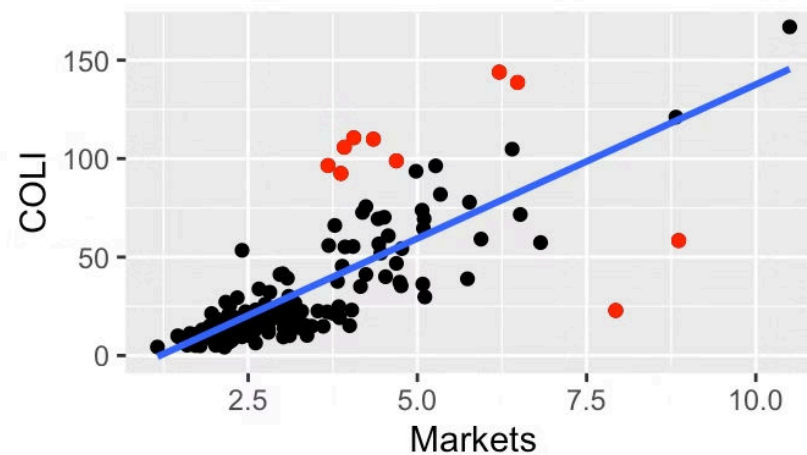
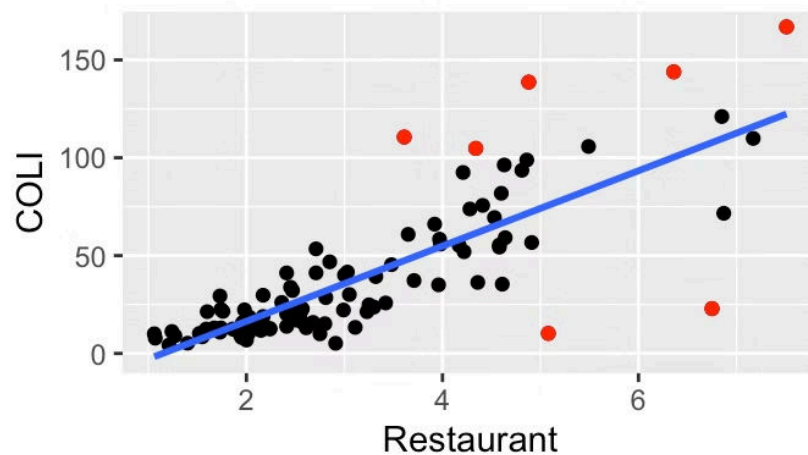
COLI and Restaurant Costs Prediction

- In both charts, there is a **positive** correlation between COLI and cost of restaurants.
- As the cost of restaurants increases, COLI also tends to increase.
- **60%** of the COLI variation explained



How's COLI related to basic factors?

- All **positive** correlation (coefficient close to 1)
- **Outliers** are acceptable (e.g local products tend to be cheaper locally)
- Countries with a higher COLI tend to have higher basic factors costs (as we expected)



How's COLI influenced by many factors

- COLI is strongly related to each factors
 - R-squared = 0.99
- Regressors are statistical significant
 - P-value < 0.05
- Residuals are scattered randomly around zero (1st chart)
- Residuals are approximately normally distributed around the line (2st chart)

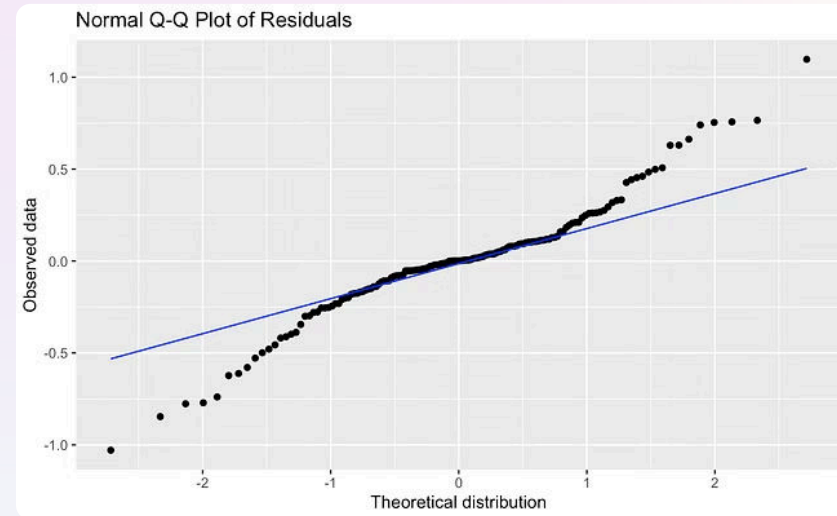
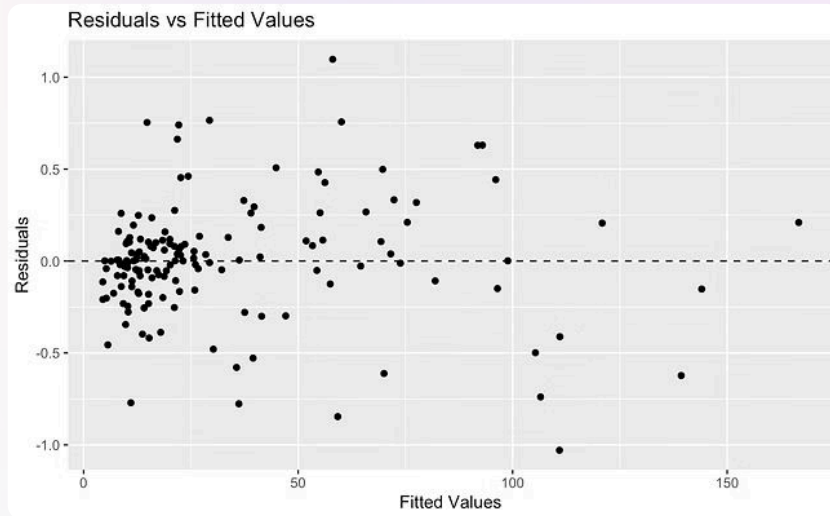
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Call:
lm(formula = COLI ~ Restaurants + Markets + MonthlyRent + Salaries,
    data = basket3)

Residuals:
    Min       1Q   Median       3Q      Max
-1.02915 -0.14226  0.00136  0.11492  1.09721

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.494e-01  7.343e-02  -2.034   0.0437 *
Restaurants  1.923e-01  3.862e-02   4.979  1.76e-06 ***
Markets      1.634e-01  3.205e-02   5.098  1.04e-06 ***
MonthlyRent  3.347e-02  8.039e-05  416.277 < 2e-16 ***
Salaries     2.508e-02  6.763e-05  370.837 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3215 on 147 degrees of freedom
Multiple R-squared:  0.9999    Adjusted R-squared:  0.9999
F-statistic: 3.645e+05 on 4 and 147 DF,  p-value: < 2.2e-16
```

How's COLI influenced by many factors



What factors most affect the COLI across countries?

Costs of restaurant

Costs of market goods

Costs of monthly rent

Salary

If there are any changes in these factors, COLI will be highly impacted.

For example, by addressing these factors cities with low-coli (e.g. Sri Lanka, Syria) would immediately improve their economic growth and quality of life .



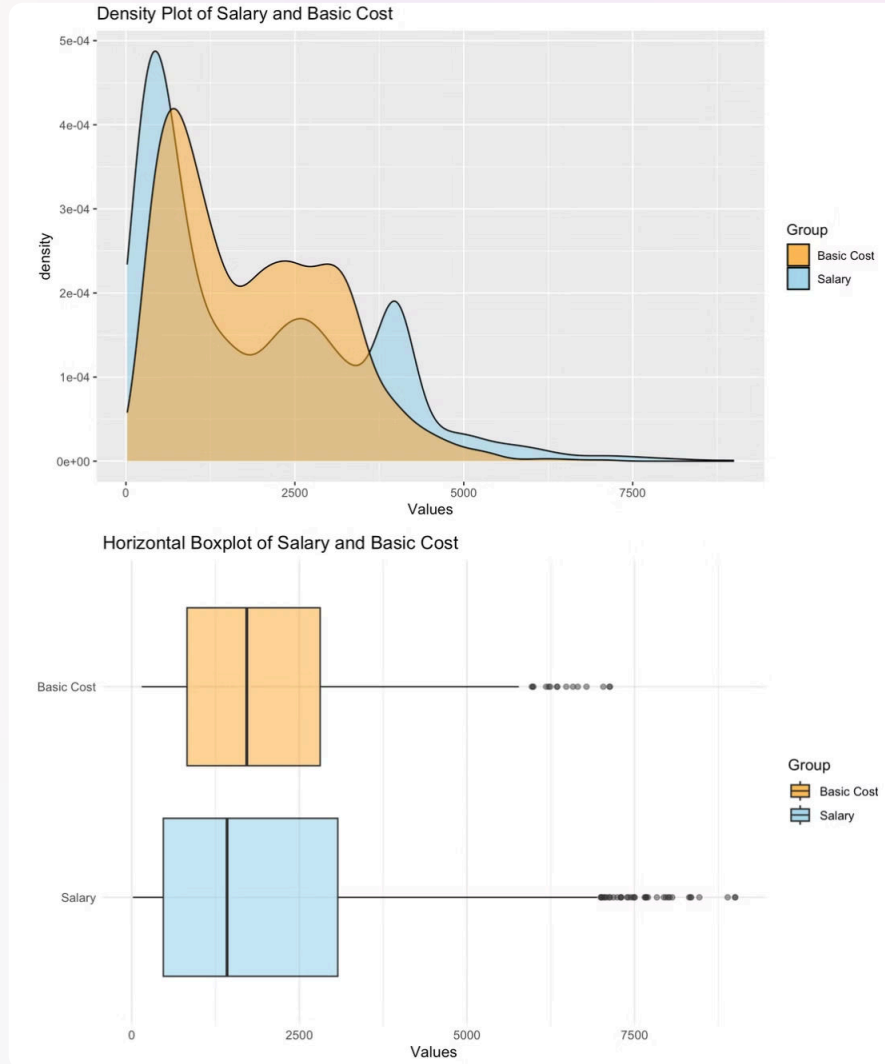
Salary vs Basic Cost

Assessing affordability levels for essential needs.

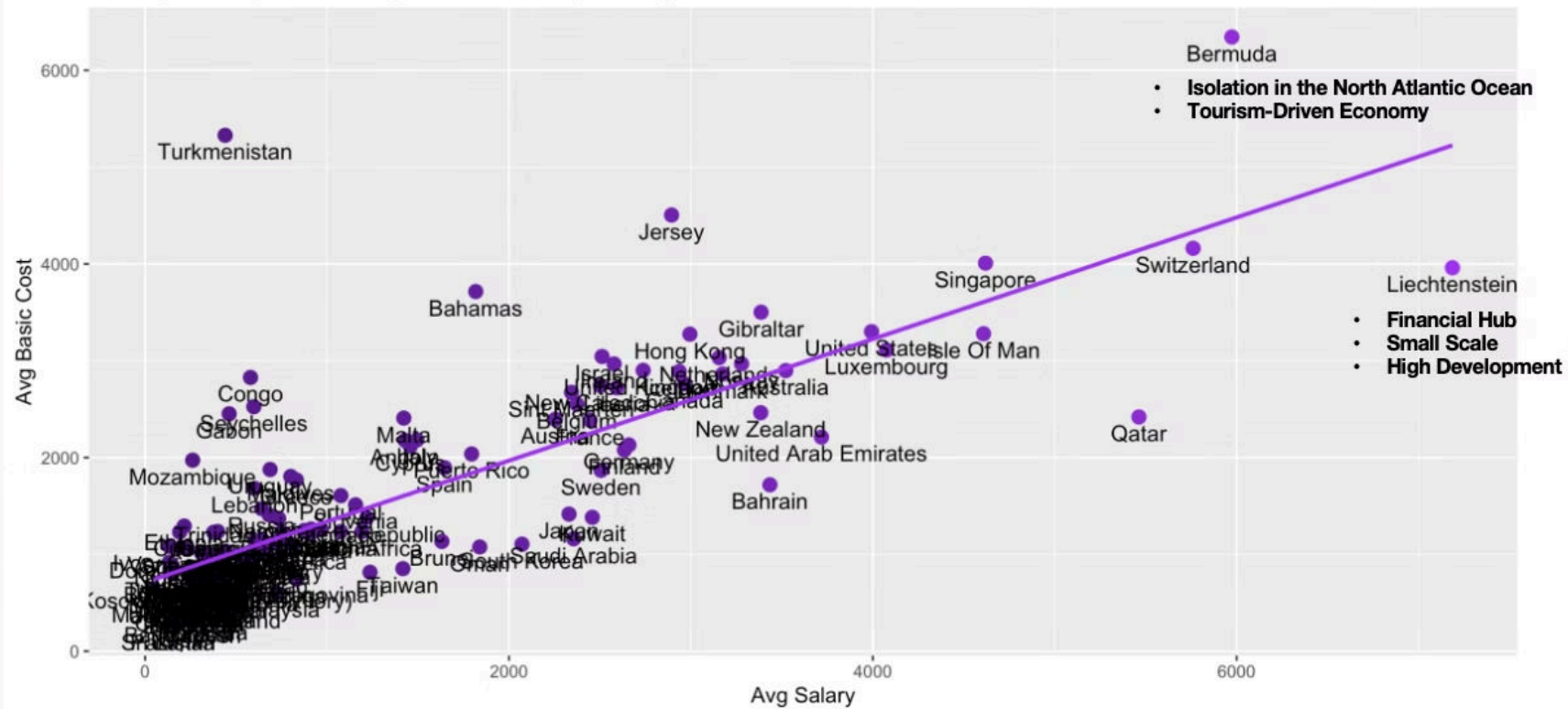
Includes everyday meals, transportation, utilities, and accommodations.

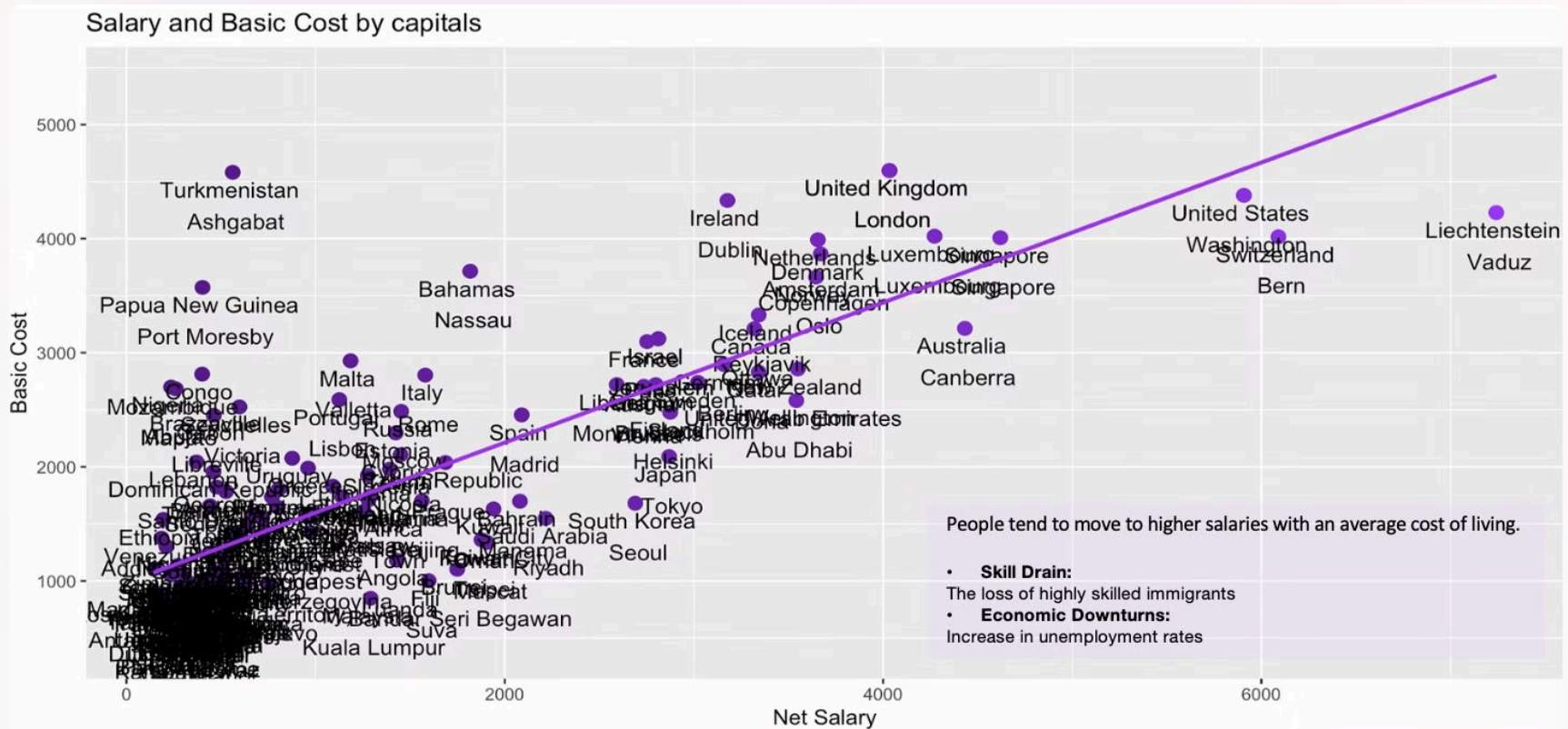


The Basic Cost compare with Salary

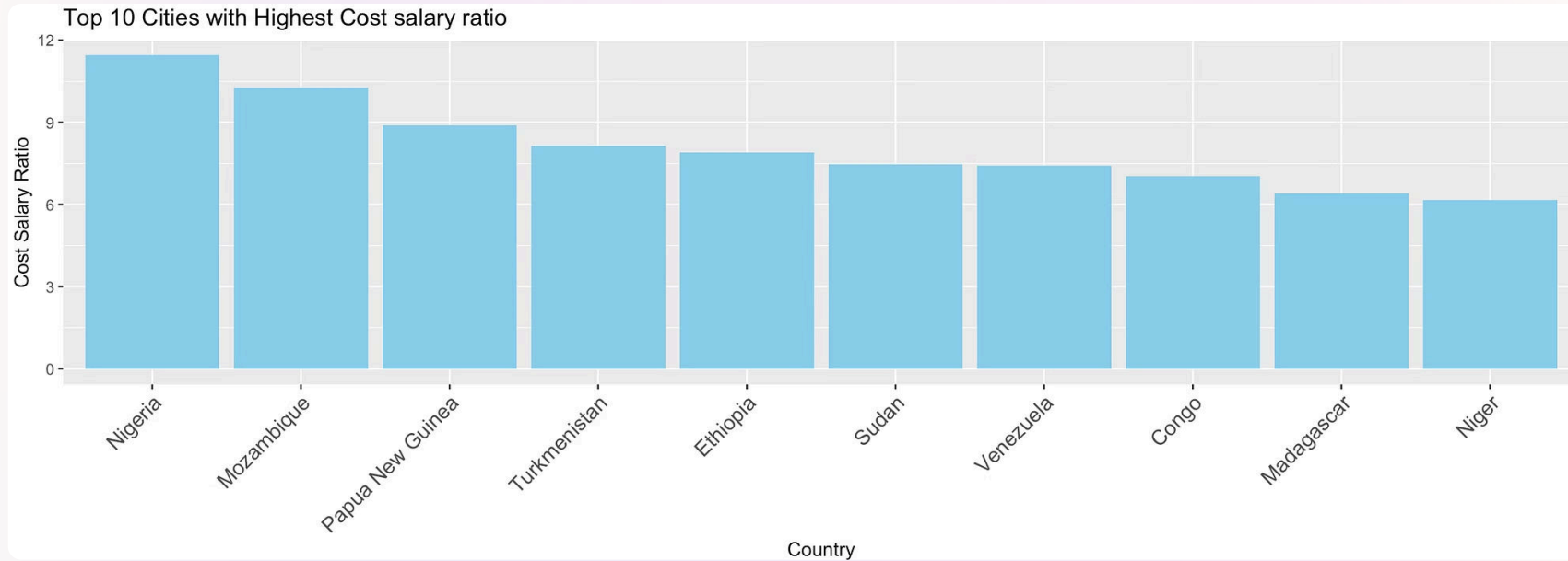


- Range of salary is **bigger** than basic cost
 - Higher volatility in salaries across cities
 - Economic conditions or industry differences
- Average of basic cost is **higher** than salary
 - Salary is not enough to cover basic living costs
 - Price levels, housing costs, inflation rates, etc.

[illegible]



Cost Salary Ratio - Highest



In some countries, the average salary is not enough to cover basic living costs.

1 Reduce the cost

1. Price controls on essential goods
2. Increase supply of housing

2 Increase salary level

1. Increase minimum wage
2. Education and training for workers



Developed VS Developing Countries

Based on [Country Insights](#) | [Human Development Reports \(undp.org\)](#) rankings.

**Very High
Development**

vhigh data

**High
Development**

high data

**Medium
Development**

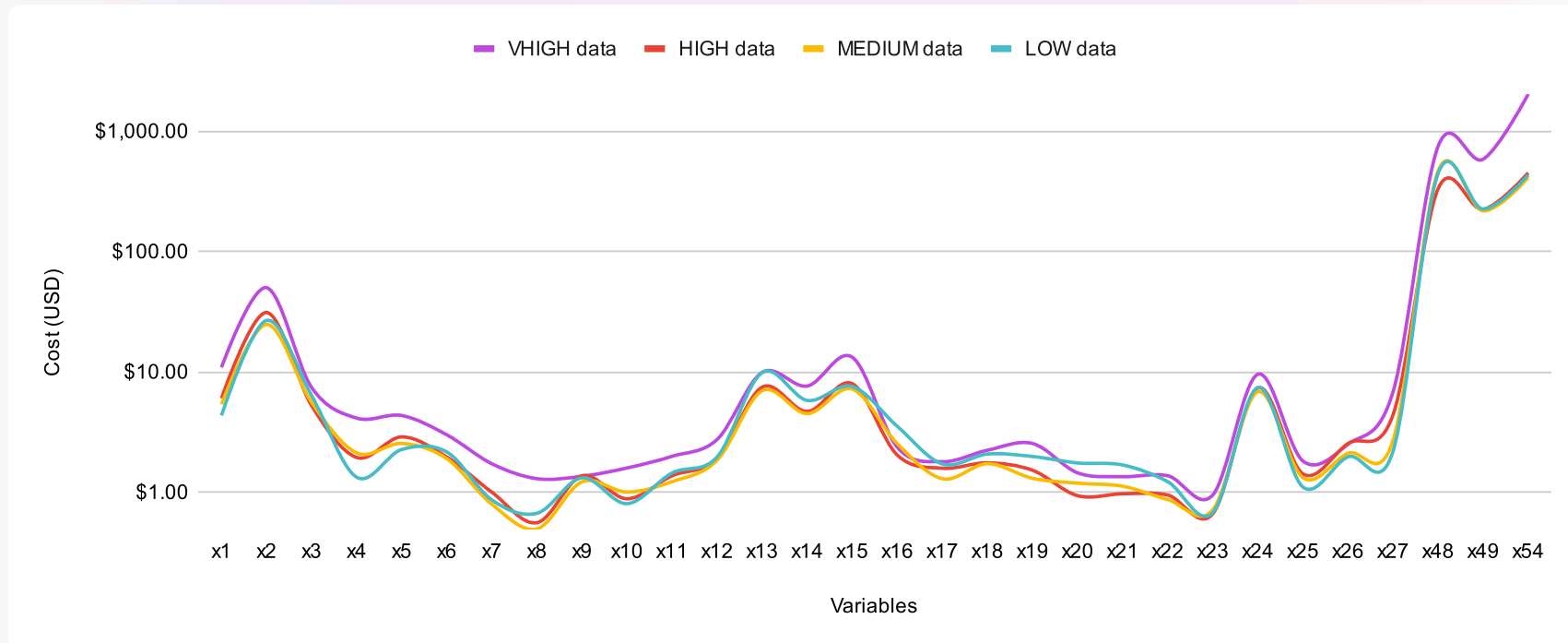
medium data

**Low
Development**

low data

Cost of living is greater in developed countries.

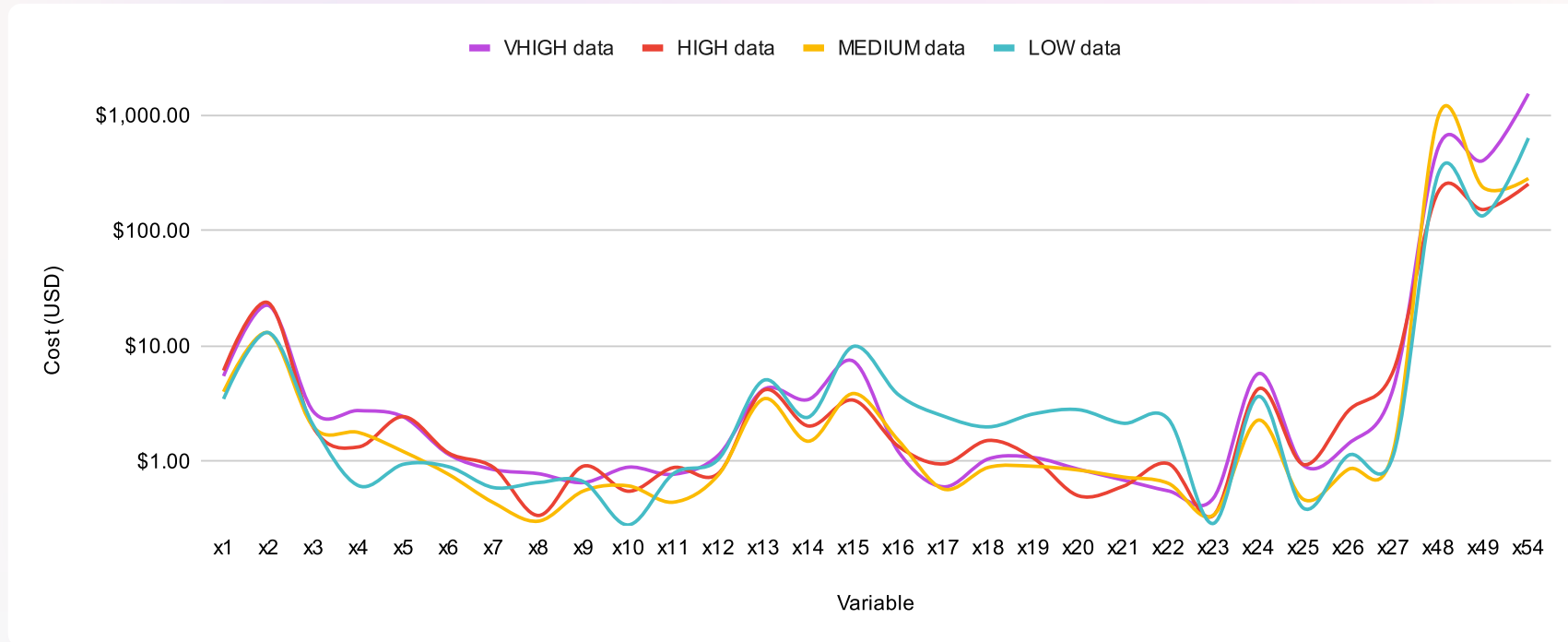
Average Variable (USD) by Country Development Level



- **Net salaries** (x54) in developed nations +4.5 times larger than closest high development countries.
- Low development countries have high average **net salary** (x54) almost equal to medium to high development countries.

Cost of living is greater in developed countries.

Standard Deviation of Variable (USD) by Country Development Level

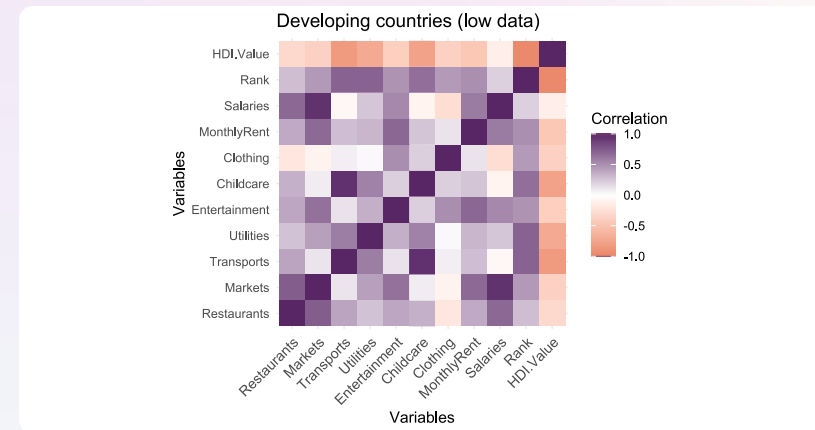
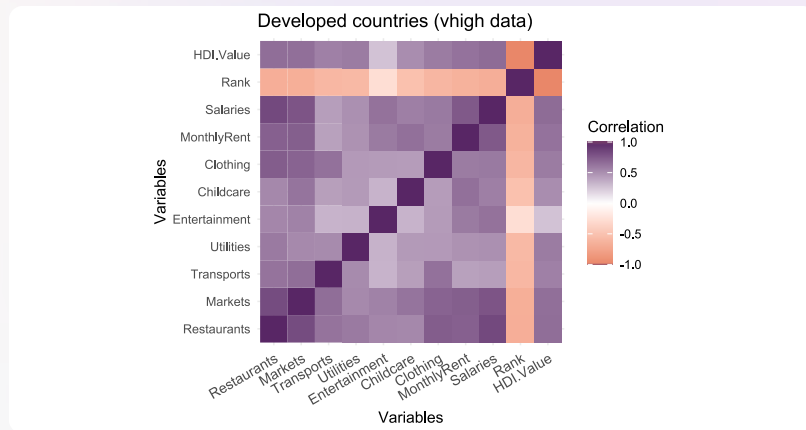


When comparing countries development level each variables show great variability in cost.

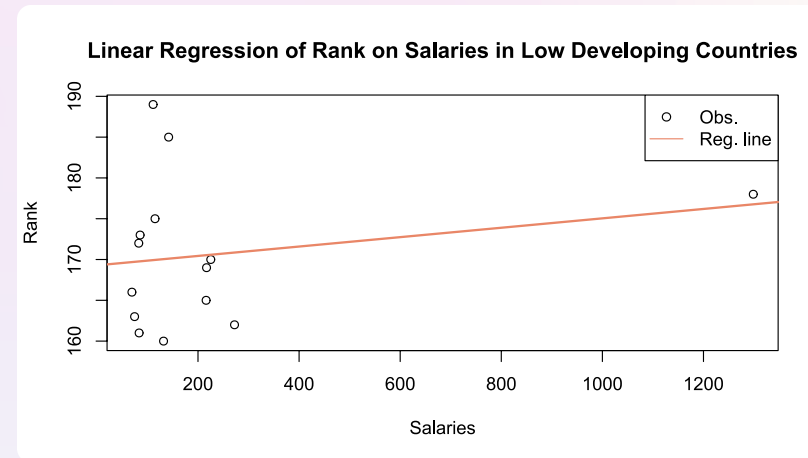
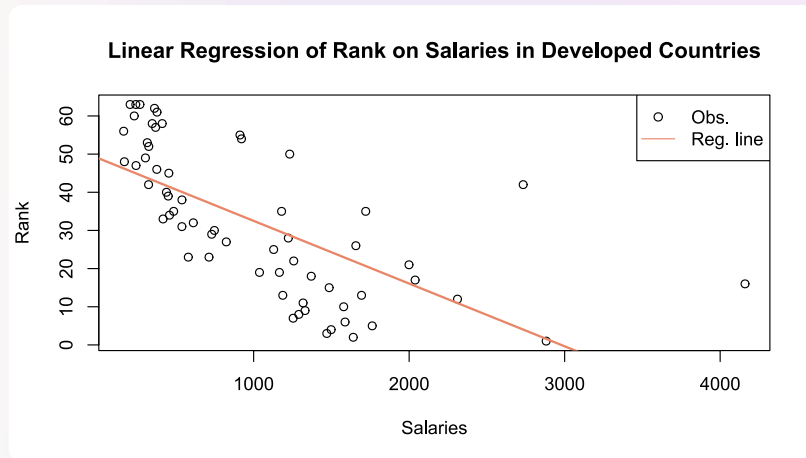
- **Net Salaries** (x54) of both developed and low development countries have higher variability compared to medium to high development countries.

Salaries Increase Proportionally with Country Development:

Correlation of Developed Countries (vhigh) and Low Development Countries (low)



Salaries Increase Proportionally with Country Development:



- In developed countries, as **net salaries** increase the country **rank number** decreases (indicating a higher country development).
- In low development countries this there is almost no linear relationship, highlighting that perhaps higher spending power isn't always equal to increased country development.

What are the main drivers of cost-of-living disparities between developed and developing nations?

There is no one overarching driver of cost-of-living disparities between developed and developing nations.

The relationship between cost-of-living and country development is complex and encompasses more than simply economic variables. To further understand the drivers of cost-of-living disparities it is necessary to look at other factors such as culture, economic structure, law and regulations, and institutional frameworks.

Thank you

Data sources:

- [Global Cost of Living \(kaggle.com\)](https://kaggle.com)
- [Country Insights | Human Development Reports \(undp.org\)](https://undp.org)

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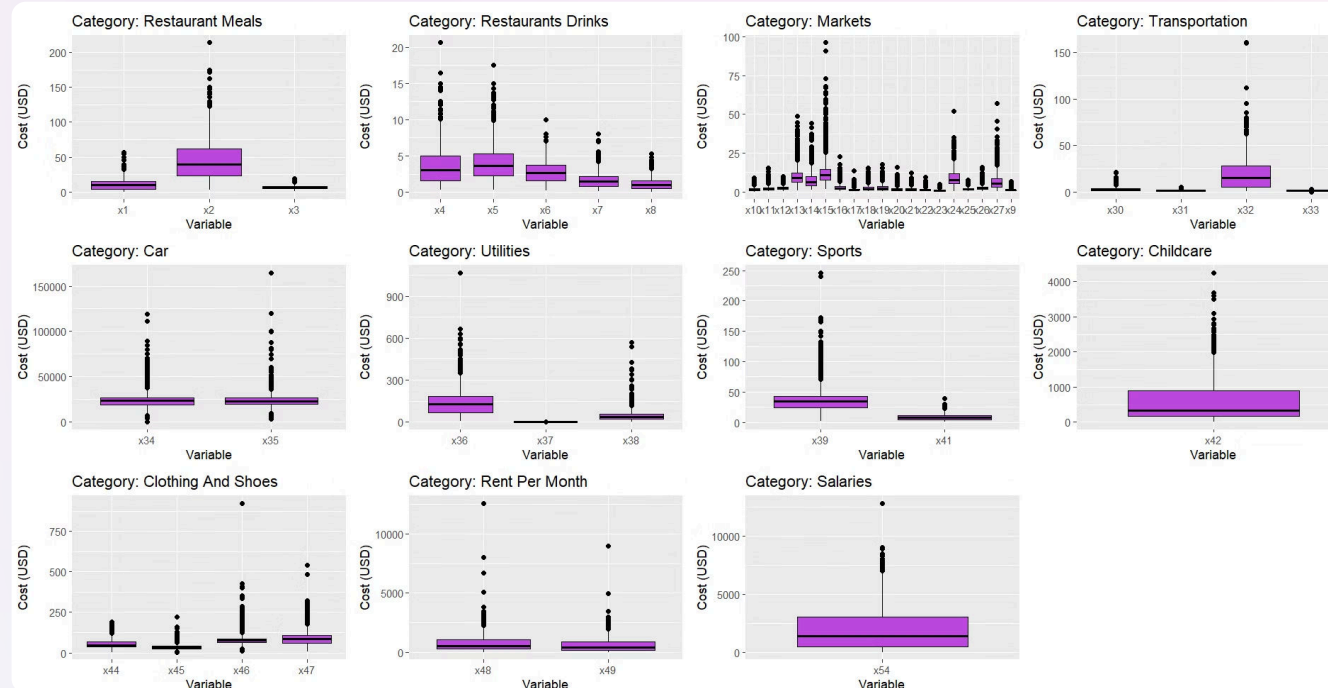
Appendix

Variables Index

Column Name	Description	Column Name	Description
city	Name of the city	x23	Water (1.5 liter bottle, at the market) (USD)
country	Name of the country	x24	Bottle of Wine (Mid-Range, at the market) (USD)
x1	Meal, Inexpensive Restaurant (USD)	x25	Domestic Beer (0.5 liter bottle, at the market) (USD)
x2	Meal for 2 People, Mid-range Restaurant, Three-course (USD)	x26	Imported Beer (0.33 liter bottle, at the market) (USD)
x3	McMeal at McDonalds (or Equivalent Combo Meal) (USD)	x27	Cigarettes 20 Pack (Marlboro) (USD)
x4	Domestic Beer (0.5 liter draught, in restaurants) (USD)	x30	Taxi Start (Normal Tariff) (USD)
x5	Imported Beer (0.33 liter bottle, in restaurants) (USD)	x31	Taxi 1km (Normal Tariff) (USD)
x6	Cappuccino (regular, in restaurants) (USD)	x32	Taxi 1hour Waiting (Normal Tariff) (U SD)
x7	Coke/Pepsi (0.33 liter bottle, in restaurants) (USD)	x33	Gasoline (1 liter) (USD)
x8	Water (0.33 liter bottle, in restaurants) (USD)	x34	Volkswagen Golf 1.4 90 KW Trendline (Or Equivalent New Car) (USD)
x9	Milk (1 liter) (USD)	x35	Toyota Corolla Sedan 1.6l 97kW Comfort (Or Equivalent New Car) (USD)
x10	Loaf of Fresh White Bread (500g) (USD)	x36	Basic (Electricity, Heating, Cooling, Water, Garbage) for 85m2 Apartment (USD)
x11	Rice (white), (1kg) (USD)	x37	1 min. of Prepaid Mobile Tariff Local (No Discounts or Plans) (USD)
x12	Eggs (regular) (12) (USD)	x38	Internet (60 Mbps or More, Unlimited Data, Cable/ADSL) (USD)
x13	Local Cheese (1kg) (USD)	x39	Fitness Club, Monthly Fee for 1 Adult (USD)
x14	Chicken Fillets (1kg) (USD)	x41	Cinema, International Release, 1 Seat (USD)
x15	Beef Round (1kg) (or Equivalent Back Leg Red Meat) (USD)	x42	Preschool (or Kindergarten), Full Day, Private, Monthly for 1 Child (USD)
x16	Apples (1kg) (USD)	x44	1 Pair of Jeans (Levis 501 Or Similar) (USD)
x17	Banana (1kg) (USD)	x45	1 Summer Dress in a Chain Store (Zara, H&M, ...) (USD)
x18	Oranges (1kg) (USD)	x46	1 Pair of Nike Running Shoes (Mid-Range) (USD)
x19	Tomato (1kg) (USD)	x47	1 Pair of Men Leather Business Shoes (USD)
x20	Potato (1kg) (USD)	x48	Apartment (1 bedroom) in City Centre (USD)
x21	Onion (1kg) (USD)	x49	Apartment (1 bedroom) Outside of Centre (USD)
x22	Lettuce (1 head) (USD)	x54	Average Monthly Net Salary (After Tax) (USD)

Global Cost of Living: EDA

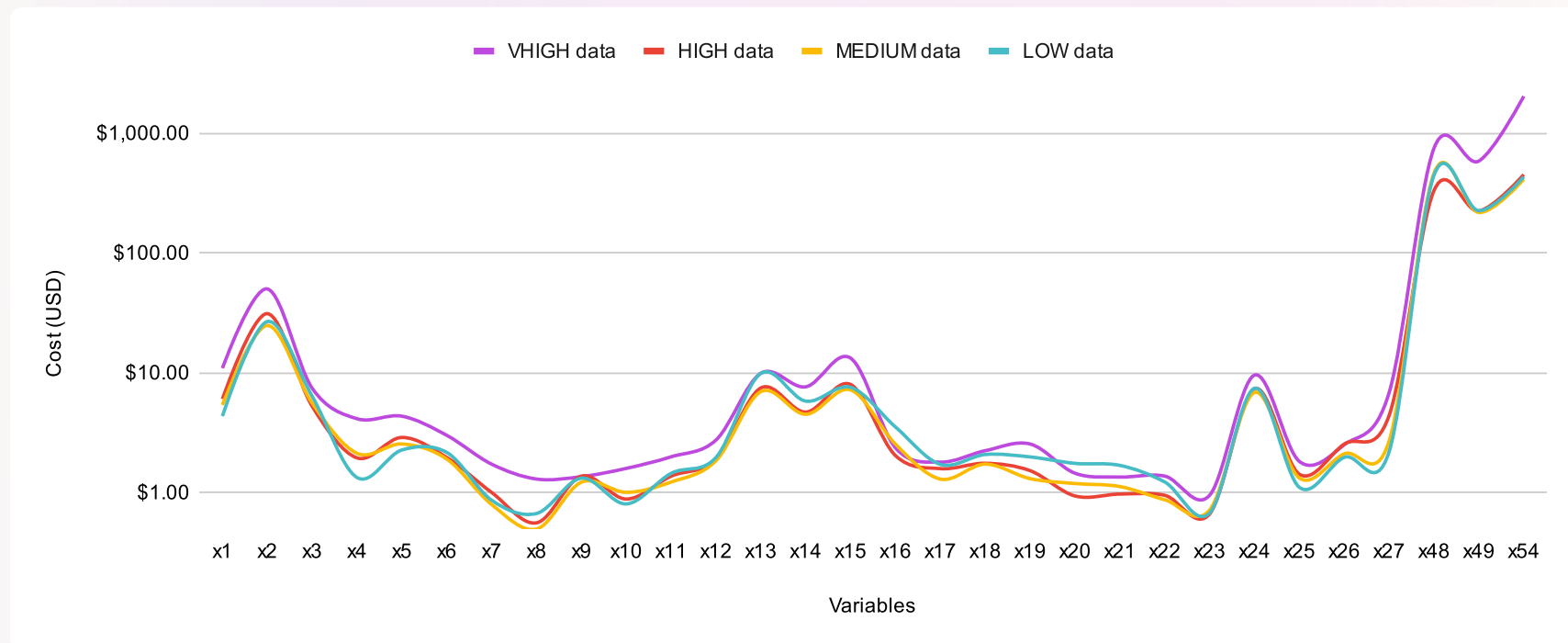
Boxplots of Variables Grouped by Category



- **Switzerland**'s higher **meal** prices (x2) and **market** prices reflect its generally elevated cost of living.
- **Qatar**'s **beer** pricing (x4) likely influenced by cultural and religious factors.
- High prices in **fitness** categories (x39, x46) may represent niche markets or specific brands.
- Outliers in **Iranian** new **car** data (x34, x35) for potential inconsistencies.

Developed VS Developing Countries

Average Variable (USD) by Country Development Level



Mean:

- **Fast food**(x3) is \$0.5 more expensive low developing nations than in medium to high development countries.
- **Cappuccinos** and **bottled water** at the restaurants(x6) are cheapest in medium to high development countries.
- **Local cheese** (x13) is expensive regardless of country development level.
- The cost of **fresh produce** (x16 to x22) also shows interesting fluctuations but overall cost of those will depend on what and how much local agriculture produces.

Standard deviation:

- Low development countries show highest variability for **fresh produce** (x16 to x22).
- Standard deviation remains consistent across all development levels for **restaurant drinks** (x4 to x8)